4 Word-Problem Translation

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RULE · Phrase-to-symbol dictionary

English phrase	Symbol
sum, total, increased by, more than (added after)	$+$
difference, decreased by	$-$
product, times, twice, double	$\times$
quotient, divided by, per	$\div$ or $\frac{a}{b}$
is, was, equals, results in, gives	$=$
a number (unknown quantity)	x, n, etc.

RULE · "Less than" reverses the order

"Eight less than n" $=\ n-8$ (not $8-n$)

Check: reword as "[number] minus [amount]." Same reversal for "subtracted from."

RULES SHEET – 2/2

RULE · Comparison phrases → inequality

English phrase	Symbol
at least, no less than	$\geq$
no more than, at most	$\leq$
more than (comparison)	$>$
less than, fewer than (comparison)	$<$

RULE · Perimeter word problems

$P = 2(\text{length} + \text{width})$

Length $= 2w + 3$, $P = 36$: $\quad 2((2w + 3) + w) = 36 \implies w = 5$

Always re-check which quantity the question asks for – width, length, or something else.